SURJECTIVITY OF A REPRESENTABLE MORPHISM OF ALGEBRAIC SPACES IS SURJECTIVITY ON THE UNDERLYING TOPOLOGICAL SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is written as a single 'Namely' paragraph; both directions of the equivalence are contained in it.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Surjectivity of a representable morphism in the functorial sense. Section section-representable, used at proof lines 819-822 and 826-829. Fixes condition (1) and its unwinding through base change to schemes.
- Description of the points of a fibre product of algebraic spaces. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian, used at proof lines 829-832. Gives surjectivity of the comparison map |T x_Y X| -> |T| x_{|Y|} |X|.
- Points of a scheme over a field and the image of a base change. used at proof lines 833-835. Recovers surjectivity of |f| by testing with T the spectrum of a field.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03H4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4553–4553. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a representable morphism of algebraic spaces over S. Then f is surjective (in the sense of Section 03HA) if and only if $|f| : |X| \to |Y|$ is surjective.*

Proof. Namely, if $f : X \to Y$ is representable, then it is surjective if and only if for every scheme T and every morphism $T \to Y$ the base change $f_T : T \times_Y X \to T$ of f is a surjective morphism of schemes, in other words, if and only if $|f_T|$ is surjective. By Properties of Spaces, Lemma 03H4 the map $|T \times_Y X| \to |T| \times_{|Y|} |X|$ is always

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surjective. Hence $|f_T| : |T \times_Y X| \to |T|$ is surjective if $|f| : |X| \to |Y|$ is surjective. Conversely, if $|f_T|$ is surjective for every $T \to Y$ as above, then by taking T to be the spectrum of a field we conclude that $|X| \to |Y|$ is surjective. $\square$